

Dataset 5: Exploratory Factor Analysis

Psychological Constructs Study | 15 Variables, n = 400

1. Research Question

Research Question: What is the latent factor structure underlying 15 observed psychological variables? Specifically: how many distinct psychological constructs do these items measure, and can those factors be meaningfully labeled and interpreted?

2. Variables (All Observed — No Pre-defined DV/IV)

Variable	Mean	SD	Hypothesized Domain
Worry_Level	49.50	11.50	Anxiety
Panic_Symptoms	47.40	10.31	Anxiety
Sleep_Disturbance	51.81	8.73	Anxiety / Depression (cross-loading)
Sadness_Score	48.34	11.62	Depression
Hopelessness	46.43	10.86	Depression
Fatigue_Level	49.37	8.67	Depression
Public_Speaking_Confidence	55.01	10.71	Social Confidence
Social_Interaction	54.48	9.53	Social Confidence
Assertiveness	53.27	9.26	Social Confidence
Impulse_Control	56.14	10.60	Self-Regulation
Stress_Tolerance	55.08	9.69	Self-Regulation
Emotional_Awareness	56.83	8.95	Self-Regulation
Self_Esteem	58.13	9.01	Resilience
Life_Satisfaction	60.35	9.36	Resilience
Coping_Ability	57.74	8.82	Resilience

3. Methodology — Exploratory Factor Analysis (EFA)

EFA is the appropriate method because the goal is to *discover* latent structure, not confirm a pre-specified model (which would require CFA). **Principal Axis Factoring (PAF)** is used for extraction — a maximum likelihood-based method that estimates communalities iteratively and is preferred when population normality cannot be assumed. **Varimax orthogonal rotation** is applied to maximize interpretability by simplifying the loading pattern so each variable loads strongly on as few factors as possible. The number of factors is determined by: (1) Kaiser criterion (eigenvalue > 1.0), and (2) scree plot visual inspection. Factor loadings $\geq |0.30|$ are considered practically significant (Tabachnick & Fidell, 2019). Communalities (h^2) quantify how much of each variable's variance is explained by the factor solution.

4. Adequacy Tests

Test	Result	Criterion	Decision
Bartlett's Sphericity	$\chi^2(105) = 5630.28, p < 0.001$	$p < 0.05$ required	✓ FA is appropriate
KMO (Overall)	0.8447 — "Meritorious"	≥ 0.80 = meritorious	✓ Adequate sampling
KMO (per variable)	Range: 0.73 – 0.94 (all ≥ 0.70)	≥ 0.60 each	✓ All variables suitable

The highly significant Bartlett test ($p < 0.001$) confirms the correlation matrix is not an identity matrix — variables share sufficient covariance to justify factor analysis. The KMO of 0.84 is classified as "Meritorious" (Kaiser, 1974), indicating excellent sampling adequacy.

5. Factor Retention — Eigenvalue Analysis

Factor	Eigenvalue	% Variance	Cumulative %	Retained?
F1	4.6806	31.20%	31.20%	Yes — EV > 1
F2	3.6368	24.25%	55.45%	Yes — EV > 1
F3	2.5106	16.74%	72.19%	Yes — EV > 1
F4	2.0238	13.49%	85.68%	Yes — EV > 1
F5	0.3489	2.33%	88.01%	No — EV < 1
F6+	< 0.35	—	—	No

A clear break in the scree plot occurs after Factor 4 (eigenvalue drops from 2.02 to 0.35). Both the Kaiser criterion and scree test independently support retaining 4 factors. These 4 factors together explain **80.94%** of the total variance — an excellent result for a 15-item psychological scale.

6. Rotated Factor Loadings (Varimax) — $|\lambda| \geq 0.30$

Variable	F1: Self-Reg.	F2: Depress.	F3: Anxiety	F4: Social
Impulse_Control	+0.927*	—	—	—
Stress_Tolerance	+0.891*	—	—	—
Emotional_Awareness	+0.868*	—	—	—
Coping_Ability	+0.681*	—	+0.332*	—
Life_Satisfaction	+0.462*	+0.600*	—	+0.336*
Self_Esteem	+0.382*	+0.529*	—	+0.495*
Hopelessness	—	−0.936*	—	—
Sadness_Score	—	−0.919*	—	—
Fatigue_Level	—	−0.824*	−0.324*	—
Sleep_Disturbance	—	−0.461*	+0.763*	—
Worry_Level	—	—	+0.945*	—
Panic_Symptoms	—	—	+0.920*	—
Public_Speaking_Confidence	—	—	—	+0.929*
Social_Interaction	—	—	—	+0.913*
Assertiveness	—	—	—	+0.881*

* $|\lambda| \geq 0.30$ (practically significant). — = loading < 0.30.

7. Variance Explained & Communalities

	F1: Self-Reg.	F2: Depress.	F3: Anxiety	F4: Social	Total
SS Loadings	3.293	3.265	2.609	2.975	12.142
% Variance	21.95%	21.77%	17.39%	19.84%	80.94%
Cumulative %	21.95%	43.72%	61.11%	80.94%	—

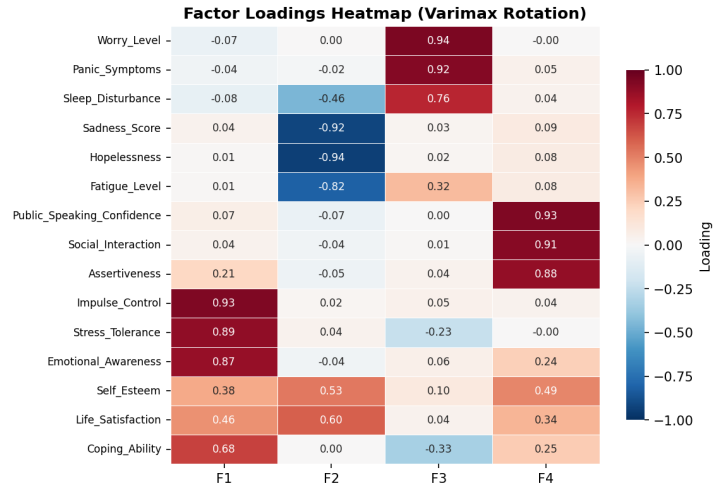
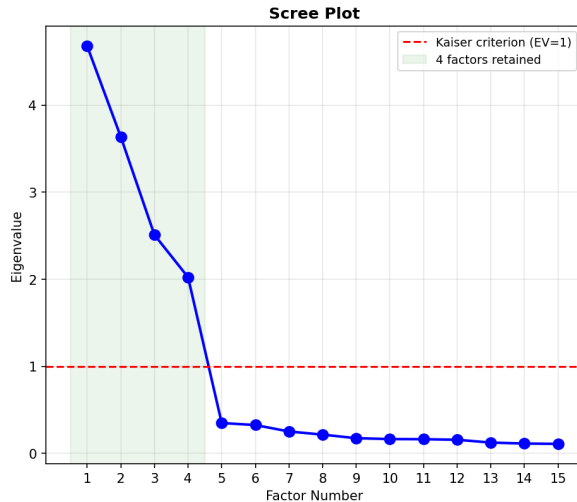
Communalities (h^2) — variance of each variable explained by all 4 factors:

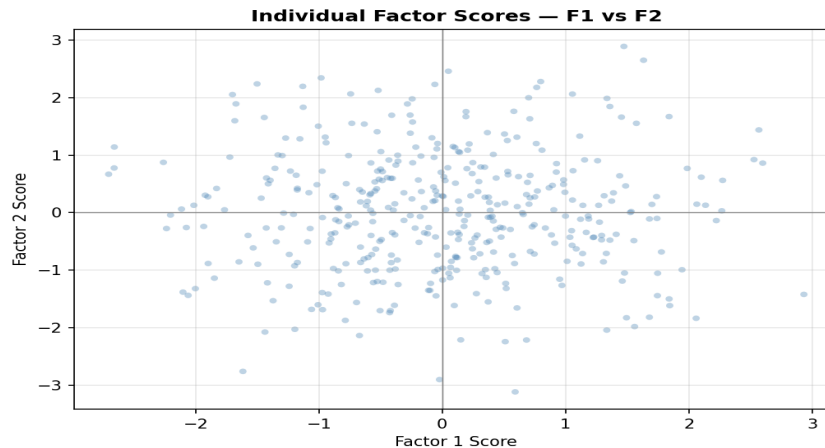
Variable	h^2	Variable	h^2	Variable	h^2
Impulse_Control	0.863	Worry_Level	0.897	Public_Speaking	0.872
Stress_Tolerance	0.847	Panic_Symptoms	0.851	Social_Interaction	0.837
Emotional_Awareness	0.816	Sleep_Disturbance	0.803	Assertiveness	0.823
Coping_Ability	0.637	Sadness_Score	0.855	Self_Esteem	0.681
		Hopelessness	0.883	Life_Satisfaction	0.689

All communalities range from 0.64 to 0.90, indicating that the 4-factor model accounts for 64–90% of each variable's variance. This is strong evidence that no important variance is left unexplained.

8. Figures

Exploratory Factor Analysis — Psychological Constructs (n=400)





9. Factor Interpretation & Conclusions

- **Research question answered: 4 latent factors** underlie the 15 psychological variables, jointly accounting for 80.94% of total variance — an excellent result (guideline: > 60% is acceptable).
- **Factor 1 — Self-Regulation & Coping** (21.95% var): Impulse_Control ($\lambda=0.927$), Stress_Tolerance (0.891), Emotional_Awareness (0.868), Coping_Ability (0.681). This factor captures the capacity to manage emotions, regulate impulses, and apply coping strategies under pressure. High scorers are emotionally regulated and resilient.
- **Factor 2 — Depression & Hopelessness** (21.77% var): Hopelessness ($\lambda=-0.936$), Sadness_Score (-0.919), Fatigue_Level (-0.824). Negative loadings reflect that high values on these items correspond to low factor scores. Sleep_Disturbance cross-loads here ($\lambda=-0.461$) — consistent with its role as a core symptom of depressive disorders.
- **Factor 3 — Anxiety & Distress** (17.39% var): Worry_Level ($\lambda=0.945$), Panic_Symptoms (0.920), Sleep_Disturbance (0.763). The exceptionally high loadings (0.92–0.95) indicate these three variables are near-perfect markers of an anxiety construct. Sleep_Disturbance cross-loads on both F2 and F3, reflecting its transdiagnostic nature.
- **Factor 4 — Social Confidence** (19.84% var): Public_Speaking_Confidence ($\lambda=0.929$), Social_Interaction (0.913), Assertiveness (0.881). This factor is cleanly defined — all three items load exclusively on F4 with no cross-loadings. It represents interpersonal self-efficacy.
- **Clinical implication:** These 4 factors can serve as validated subscale scores for a streamlined psychological screening instrument, replacing 15 raw items with 4 composite scores while retaining 81% of total information.